Proposal: Optimizing Mosquito Repellent Dispersion Utilizing Autonomous Drone Controls

Group 2: Joe Conroy, Noah Constable

Problem Statement

Experiment: our intent is to compare up to two methods of utilizing autonomous drones to dispense mosquito repellent in a marshland setting. Contrasting with the current adopted method of using manually controlled drones, our goal is to determine if it is more efficient to instead disperse the repellent using either: a truck that can operate the drone(s) autonomously with a real-world model; or a collection of drones with edge processing driven by computer vision.

Survey: our intent is to identify spatial data analysis tools and techniques that can be used to optimize mosquito control deployments. This can range from applications for autonomous drones to tools to identify the optimal areas and time windows to deploy controls.

Depending on our findings for the experiment proposal, we may only choose one approach to experiment, two methods to contrast them, or do the survey instead.

Introduction

Automated analysis and prediction model generation has advanced quite considerably to present day. Notable examples range from Waymo's autonomous ridesharing vehicles [1], John Deere's Precision AG Technology [2], and the recent application of attack drones in Operation Spider's Web [3]. All three enterprises are also examples of spatial data being used to remove people from harm's way, reduce costs or improve production. Presuming that the hardware for data training remains readily accessible, applying spatial data will continue to provide economic and societal benefits.

One area where this technology is likely going to become critical is environmental preservation and adaptation. The most recent (2022) report of the IPCC indicates that various natural and developed ecosystems are all subject to adverse impacts from the changes in the climate caused by greenhouse gas emissions from human enterprises should the global average temperature reach 1.5^0C above the global average in the pre-industrial era [6]. You can observe Figure SPM.2, which was taken directly from the IPCC's Sixth Assessment Report, on the next page to see how various systems are expected to be affected by climate change.

One significant expected trend to note of is that terrestrial ecosystems across the globe are highly likely to experience attribution from climate change. In addition, the North American, European, Small Island and Arctic regions are all highly likely to experience attributions from climate change. It is also clear that nearly all cities, infrastructure, health and (human) wellbeing are expected to experience adverse impacts without notable positives. Based on these trends, should they come to pass, we can expect that managing these risks will involve deciding whether to prioritize human systems or ecosystems.

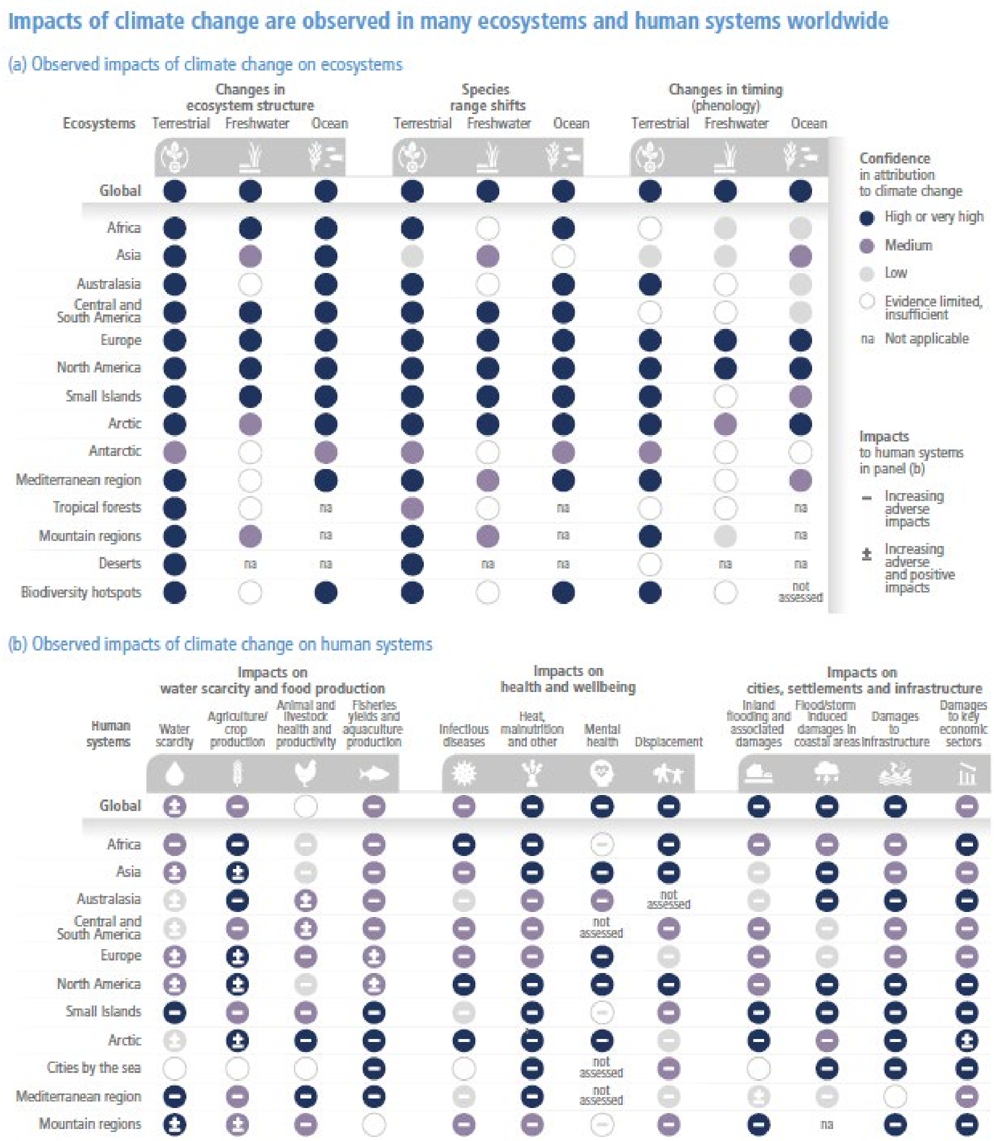

Figure SPM.2 | Observed global and regional impacts on ecosystems and human systems attributed to climate change. Confidence levels reflect uncertainty in attribution of the observed impact to climate change. Global assessments focus on large studies, multi-species, meta-analyses and large reviews. For that reason they can be assessed with higher confidence than regional studies, which may often rely on smaller studies that have more limited data. Regional assessments consider evidence on impacts across an entire region and do not focus on any country in particular.

Three years after the IPCC Sixth Report, studies had reported that we were likely to pass the 1.5^{0}C threshold within a 20-year period [7] and that global temperature trends over a consecutive 12-month period are indicative of being on track to passing 1.5^{0}C [8]. Assuming the assessments from these studies were correct, resource management is going to become critical in the coming decades. Human resources are particularly going to become increasingly strained for availability, as expertise and adaptability will be indispensable for handling multiple crises of different and overlapping varieties. Automation will be necessary to both free resources up and fill in gaps where people are not available to help.

A particular example of environmental control that interests us and will likely be affected by climate change is managing mosquito populations. Mosquitos can carry deadly diseases such as Dengue, Malaria, West Nile and Zika viruses [4]. Controlling mosquito populations is thus a critical line of defense for preventing deadly diseases from spreading to people. To control mosquito populations, the Minneapolis Mosquito Control District (MMDC) employs various larval controls such as Bacillus thuringiensis isrealensis (Bti), methoprene and Spinosad [9]. Deploying these controls requires identifying mosquito nesting locations while the mosquitos are still in their larval state.

There are three main challenges to overcome when reducing mosquito populations with the MMCD's current method. Exposure to the mosquito population greatly increases the chances of mosquito bites and in turn the risk of transmitting diseases. At the same time, attempting to control mosquito populations can disturb the freshwater food webs they tend to be a part of if deployment isn't methodical. For example, Bti presents risks to chironomid larvae as well as to mosquito larvae [5]. Chironomids tend to be diverse, highly digestible, and protein-rich sources of food for various freshwater creatures. Deploying too much Bti risks significantly reducing the chironomid species and thus doing more harm to the environment. The third challenge is the required accuracy of deployment procedure itself, and that is where we believe our method will be able to fill a gap caused by resource constraints. Deployment can only be done on a seasonal basis to target the larvae before they develop into adults. A successful deployment framework for mosquito controls will thus be deployed over a familiar seasonal landscape, with accurate identification of the targeted areas and a measured volume of the control deployed.

Literature Review

[Include sources for methods to study here, including edge processing and real-world models]

Limitations of Existing Work

The current method of mosquito control involves using a manually piloted drone with a camera feed for the human pilot to identify and deploy the mosquito control method of choice. While this does prevent the person from being too exposed to the mosquito swarm, the methodology…

Meanwhile the existing edge processing and real-world models tend to not be focused on a task as specific as identifying mosquito populations or nesting areas…

Methodology

Should we find an ideal or obvious spatial learning method for the drones to adopt, then we shall conduct a simulated experiment in which the drones need to correctly identify targets for deploying the anti-mosquito control, approaching the target, and deploying the control method on said target in an appropriate manner. The simulations will also be timed based on an approximation of the drone's battery life, with time depleting based on the actions the drones take, how far they travel, and at what elevation do they hover.

Evaluating each method's success rate will be based on multiple factors. The number of false positives for deploying the control can inform us about the likelihood of harming the environment. A failed deployment can be counted as either deploying too much of the control or missing the target. True negatives occur when a population is present but never identified, targeted or deployed against within the time limit. Any of these instances are indicative of a waste of resources. Measuring both false positives and true negatives will enable us to contrast the expected failure type.

Resource management can also be evaluated based on what was required to train a model to achieve acceptable accuracy and efficient use of time, as well as how many resources were required of the agents making the decisions. This measurement will help clarify if the preferred method involves treating drones as remotely piloted devices (our baseline), as edge devices (drones make decisions on their own), or as an extension of a platform hosting a trained model. Furthermore, we can determine if a portable computer, a computer-embedded vehicle, or a remote data center is a preferred host for the trained model.

Schedule

Our project milestones are listed below:

Date	Milestone (Experiment Option)	Milestone (Survey Option)
3/12	Decide to Either Experiment with One or Two Methods for Mosquito Control, or to Survey Potential Methods Instead	
3/15	Obtain Data for Generating Simulated Environment(s)	List All Potential Data That Can Be Utilized for Training
3/20	Construct Algorithm(s) to Generate Test (and Training) Environments, and Algorithm(s) to Train Drone Decision-Making	List All Potential Algorithms We Found That Can Utilize the Data from 3/15 for Our Intended Goal

3/25	Coded Algorithm(s) For Training Mosquito Seeking Model(s)	Write Down Strengths and Limitations for the Algorithms Written Down on 3/20
3/31	Successfully Train Mosquito-Seeking Model(s) And Complete Tests	Cutoff Date for All Exploration of Alternative Methods
4/7	Analysis of Model Performance, Resource Demands Written Down	Write Down our Conclusion From Our Survey
4/14	Paper Rough Draft Due	
4/20	Paper Revisions Following Draft Feedback	
4/21-4/30	Project Presentation	
5/8	Final Paper Due	

Resources

1. WAYMO; *WAYMO Home Page*; 2019-2026 Waymo LLC; *https://waymo.com/*
2. John Deere; *Prevision AG Technology Page*; 2026 Deere & Company; https://www.deere.com/en/technology-products/precision-ag-technology/
3. CSIS; *How Ukraine's Operation "Spider's Web" Redefines Asymmetric Warfare*; 2025 Center for Strategic & International Studies; https://www.csis.org/analysis/how-ukraines-spider-web-operation-redefines-asymmetric-warfare
4. Texas Health and Human Services, 2026 Texas Department of State Health Services, *List of Mosquito-Borne Diseases*, https://www.dshs.texas.gov/mosquito-borne-diseases/list-mosquito-borne-diseases
5. Stefanie Allgeier, Anna Friedrich, Carsten A. Brühl, *Mosquito control based on Bacillus thuringiensis israelensis (Bti) interrupts artificial wetland food chains*, Science of The Total Environment, Volume 686, 2019, Pages 1173-1184, ISSN 0048-9697, https://doi.org/10.1016/j.scitotenv.2019.05.358.
6. IPCC; 2022 Intergovernmental Panel on Climate Change; *Climate Change 2022: Impacts, Adaptation and Vulnerability; Working Group II Contribution to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change*; https://www.ipcc.ch/report/ar6/wg2/
7. Bevacqua, E., Schleussner, CF. & Zscheischler, J. A year above 1.5 °C signals that Earth is most probably within the 20-year period that will reach the Paris Agreement limit. *Nat. Clim. Chang.* **15**, 262–265 (2025). https://doi.org/10.1038/s41558-025-02246-9
8. Cannon, A.J. Twelve months at 1.5 °C signals earlier than expected breach of Paris Agreement threshold. *Nat. Clim. Chang.* **15**, 266–269 (2025). https://doi.org/10.1038/s41558-025-02247-8

9. MMCD, 2026, *Mosquito Control*, https://mmcd.org/mosquito-control/
10.